Periodic Trends

Subject & topics: Chemistry | Inorganic | Periodic Table **Stage & difficulty:** A Level P1

Part A

Melting points of third row elements

Which graph best shows the variation of melting point of the third row elements?

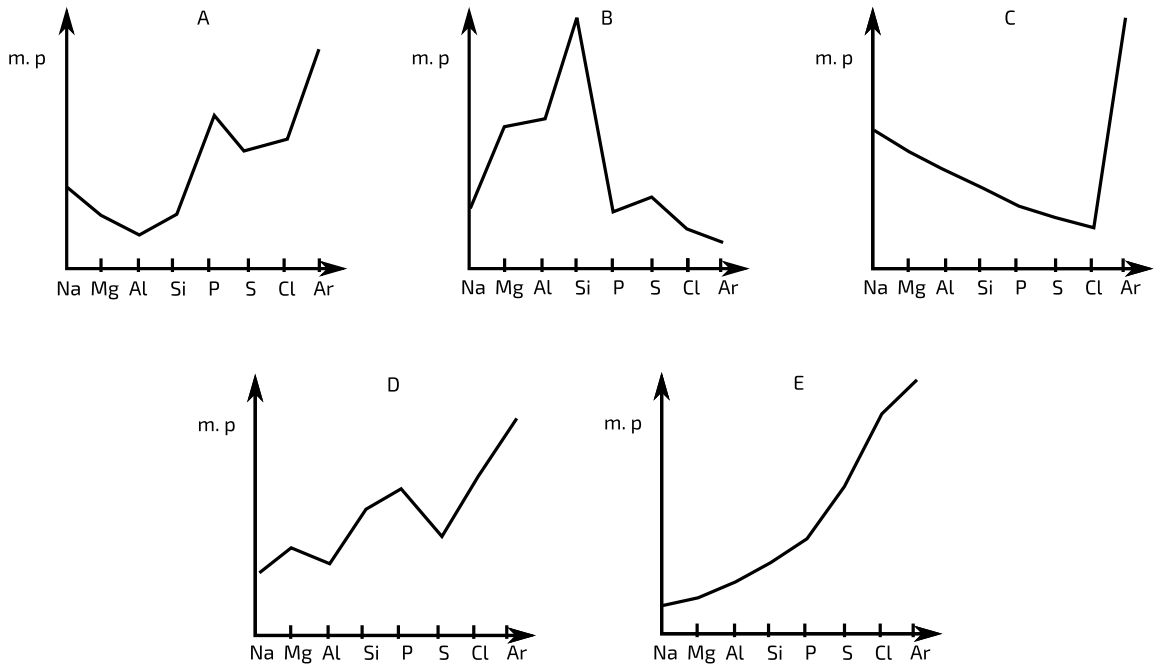

Figure 1: Melting points across 3rd period

- A
- B
- C
- D

Part B

Trends in halogens

Which graph correctly describes a trend found in the halogen group?

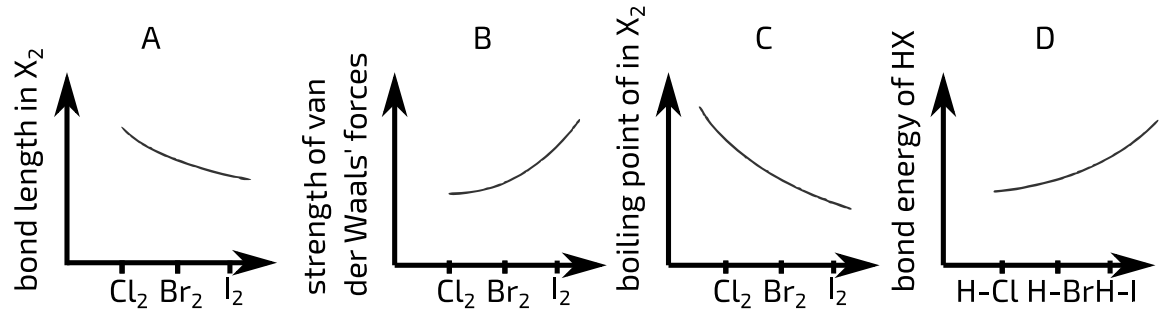

Figure 2: Trends in the halogen group

- A
- B
- C
- D

Part A adapted with permission from UCLES, A-Level Chemistry, June 1991, Paper 1, Question 17;
Part B adapted with permission from UCLES, A-Level Chemistry, June 1995, Paper 4, Question 15

Question deck:
STEM SMART Chemistry Week 9

Sizes of Atoms and Ions

Subject & topics: Chemistry | Inorganic | Periodic Table **Stage & difficulty:** A Level P1

Part A

Sizes of ions

Which of the following sets of diagrams best indicates the relative radii of the atom and most common ion of sodium and chlorine?

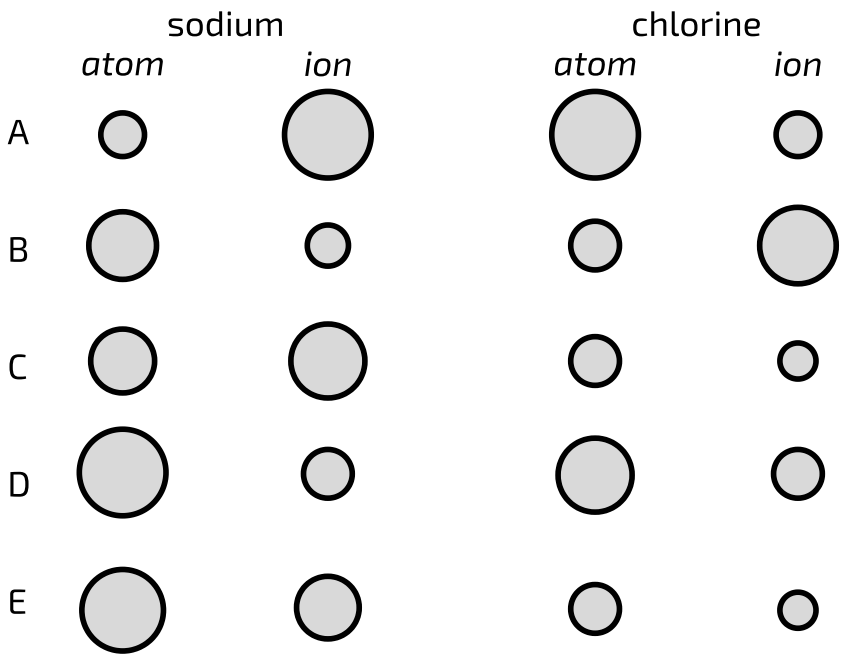

Figure 1: Sizes of Na and Cl atoms and ions

- A
- B
- C
- D
- E

Part B

Largest radius

Which species represented by the following formulae has the largest radius?

Part A adapted with permission from UCLES, A-Level Chemistry, November 1991, Paper 1, Question 15;
Part B adapted with permission from UCLES, A-Level Chemistry, November 1994, Paper 1, Question 12

Question deck:
STEM SMART Chemistry Week 9

Heating Calcium Hydroxide

Subject & topics: Chemistry | Inorganic | Periodic Table **Stage & difficulty:** A Level P1

Part A

Heating calcium hydroxide

Write the equation for the action of heat on calcium hydroxide, including state symbols, balancing the equation with the lowest possible stoichiometric coefficients.

Part B

Decomposition of calcium hydroxide

Which of the following explains why magnesium hydroxide decomposes at a lower temperature than calcium hydroxide?

1. MgO has a larger magnitude lattice energy than CaO.
2. Mg has higher first and second ionisation energies than Ca.
3. $\mathrm{Mg(OH)_2}$ has a larger magnitude lattice energy than $\mathrm{Ca(OH)_2}$.

- **1, 2** and **3** are correct
- **1** and **2** only are correct
- **2** and **3** only are correct
- **1** only is correct
- **3** only is correct

Part A,B adapted with permission from UCLES, A-Level Chemistry, 1989, Paper 2, Question 1;
Part C adapted with permission from UCLES, A-Level Chemistry, 1988, Paper 3, Question 15

Question deck:

STEM SMART Chemistry Week 9

Silver Ions

Subject & topics: Chemistry | Inorganic | Periodic Table **Stage & difficulty:** A Level P1

An aqueous solution containing Br^- ions is treated with $AgNO_3\,(aq)$, giving a precipitate **P** which is then tested for its solubility in concentrated $NH_3\,(aq)$.

What is the colour of **P** and its solubility in $NH_3\,(aq)$?

	colour of P	**solubility in $NH_3\,(aq)$**
A	white	insoluble
B	white	slightly soluble
C	cream	slightly soluble
D	yellow	insoluble

- A
- B
- C
- D

Adapted with permission from UCLES, A-Level Chemistry, June 1996, Paper 3, Question 17

Question deck:
STEM SMART Chemistry Week 9

Group 2

Subject & topics: Chemistry | Inorganic | Periodic Table **Stage & difficulty:** A Level P1

Part A

Precipitates

Which pair of $0.1\,\mathrm{mol\,dm^{-3}}$ aqueous solutions is most likely to give a precipitate when added together?

- KBr and $\mathrm{MgSO_4}$
- $\mathrm{MgSO_4}$ and $\mathrm{SrCl_2}$
- $\mathrm{NH_3}$ and $\mathrm{CaCl_2}$
- $\mathrm{NaNO_3}$ and $\mathrm{BaCl_2}$

Part B

Properties of Group 2 elements

Which of the following is a property of the elements in Group 2, magnesium to barium?

- They all have outer electronic structures $\mathrm{ns^2\ np^2}$
- They all react explosively with cold water liberating hydrogen
- They all form covalent chlorides $\mathrm{MCl_2}$
- They all form oxides MO
- They all liberate chlorine from concentrated hydrochloric acid

Part A adapted with permission from UCLES, A-Level Chemistry, June 1996, Paper 3, Question 14
Part B adapted with permission from OCSEB, A-Level Chemistry, June 1995, Paper 1, Question 19

Question deck:
STEM SMART Chemistry Week 9

Groups and Electronegativity

Subject & topics: Chemistry | Inorganic | Periodic Table **Stage & difficulty:** A Level P1

Part A

Electronegativity

In the periodic table, the electronegativity of the elements in:

1. Period 3 increases from sodium to chlorine.
2. Group 2 increases from barium to beryllium.
3. Group 7 increases from iodine to fluorine.

- **1, 2** and **3** are correct
- **1** and **2** only are correct
- **2** and **3** only are correct
- **1** only is correct
- **3** only is correct

Part B

Groups

Which of the following statements describing the characteristics of elements within any one group of the Periodic Table are correct?

1. The elements are either all metals or non-metals.
2. The melting points of the elements increase with increasing proton (atomic) number.
3. The first ionisation energies of the elements generally decrease with increasing proton (atomic) number.

- **1, 2** and **3** are correct
- **1** and **2** only are correct
- **2** and **3** only are correct
- **1** only is correct
- **3** only is correct

Part A adapted with permission from UCLES, A-Level Chemistry, June 1991, Paper 1, Question 35;
Part B adapted with permission from UCLES, A-Level Chemistry, November 1993, Paper 4, Question 34

Question deck:
STEM SMART Chemistry Week 9

Group 2 Sulfates

Subject & topics: Chemistry | Inorganic | Periodic Table **Stage & difficulty:** A Level P1

Part A

Sulfuric acid and barium hydroxide

Dilute sulfuric acid was added to aqueous barium hydroxide until the acid was in excess. Which graph shows the variation in the total number of ions in solution?

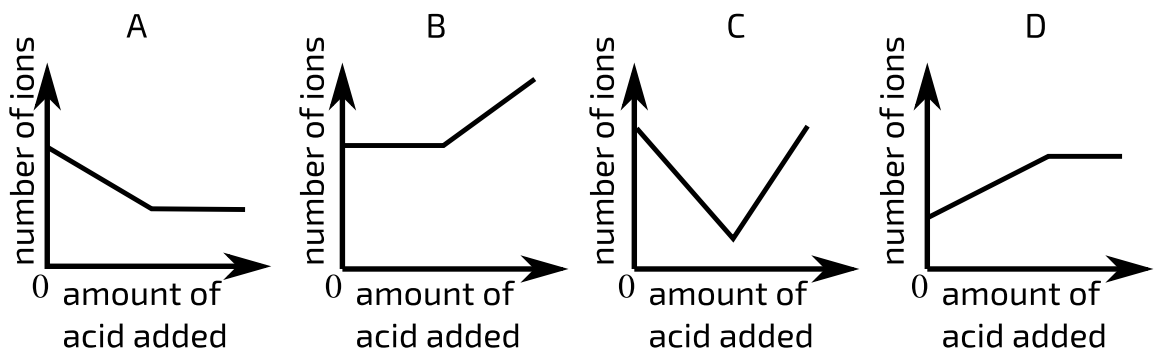

Figure 1: number of ions vs. amount of acid added

- A
- B
- C
- D

Part B

Group 2 sulfates

What changes occur in the **magnitudes** of:

(i) the lattice energy,
(ii) the enthalpy of hydration,
(iii) the solubility of the sulfates,

as Group 2 is descended?

	lattice energy	enthalpy change of hydration	solubility of sulfate
A	decrease	decrease	decrease
B	decrease	increase	decrease
C	increase	decrease	increase
D	decrease	increase	increase

- A
- B
- C
- D

Part A adapted with permission from UCLES, A-Level Chemistry, June 1994, Paper 4, Question 16;
Part B adapted with permission from UCLES, A-Level Chemistry, June 1995, Paper 4, Question 13

Question deck:
STEM SMART Chemistry Week 9

Atomic and Ionic Radii and Ionisation Energy 1

Essential Pre-Uni Chemistry D3.1

Subject & topics: Chemistry | Inorganic | Periodic Table **Stage & difficulty:** A Level P1

There are trends evident in atomic and ionic radii. Ionisation energies also show trends. Complete the sentences below with the words 'increase' or 'decrease', to indicate what happens to the radii and ionisation energy of the atoms or ions [(a)–(f)], or to the ionisation energies [(g)–(i)].

Part A

Along a period, L-R

Going along a period from left to right, the atomic radii...

- decrease
- increase

Part B

Down a group

Going down a group, the atomic radii...

- decrease
- increase

Part C

Electrons removed

As successive electrons are removed from the same atom/ion, the radii...

- decrease
- increase

Part D

Same charge, down a group

The radii of ions of the same charge, on descending a group...

- increase
- decrease

Part E

Adding electrons

As successive electrons are added to one atom to make increasingly negative ions, the radii...

- increase
- decrease

Part F

Along period, L-R

Along a period from left to right, the radii of isoelectronic species generally...

- increase
- decrease

Part G

Along period, L-R

Along a period from left to right, the first ionisation energies generally...

- increase
- decrease

Part H

Down a group

Going down a group, the first ionisation energies...

- increase
- decrease

Part I

Ionisation energies

Successive ionisation energies for the same element...

- decrease
- increase

Question deck:

STEM SMART Chemistry Week 9

Atomic and Ionic Radii and Ionisation Energy 2

Essential Pre-Uni Chemistry D3.2

Subject & topics: Chemistry | Inorganic | Periodic Table **Stage & difficulty:** A Level P2

Part A

$[\mathrm{Na\ Mg\ Al}]$

Which would have the smallest radius in the set $[\mathrm{Na\ \ Mg\ \ Al}]$?

- Mg
- Na
- Al

Part B

$[\mathrm{Na^+\ Mg^{2+}\ Al^{3+}}]$

Which would have the smallest radius in the set $[\mathrm{Na^+\ \ Mg^{2+}\ \ Al^{3+}}]$?

- $\mathrm{Mg^{2+}}$
- $\mathrm{Al^{3+}}$
- $\mathrm{Na^+}$

Part C

$[\mathrm{B\ Al\ Ga\ In\ Tl}]$

Which would have the smallest radius in the set $[\mathrm{B\ \ Al\ \ Ga\ \ In\ \ Tl}]$?

- B
- Al
- Ga
- Tl
- In

Part D

$[\mathrm{Si^{4-}\ P^{3-}\ S^{2-}\ Cl^{-}}]$

Which would have the largest radius in the set $[\mathrm{Si^{4-}\ \ P^{3-}\ \ S^{2-}\ \ Cl^{-}}]$?

- $\mathrm{P^{3-}}$
- $\mathrm{S^{2-}}$
- $\mathrm{Cl^{-}}$
- $\mathrm{Si^{4-}}$

Part E

$[Ti^{4+}\ Zr^{4+}\ Hf^{4+}\ Rf^{4+}]$

Which would have the smallest radius in the set $[Ti^{4+}\ \ Zr^{4+}\ \ Hf^{4+}\ \ Rf^{4+}]$?

- Ti^{4+}
- Rf^{4+}
- Hf^{4+}
- Zr^{4+}

Part F

$[Fe\ Fe^{2+}\ Fe^{3+}\ Fe^{2-}]$

Which would have the largest radius in the set $[Fe\ \ Fe^{2+}\ \ Fe^{3+}\ \ Fe^{2-}]$?

- Fe^{3+}
- Fe^{2-}
- Fe
- Fe^{2+}

Question deck:

STEM SMART Chemistry Week 9

Atomic and Ionic Radii and Ionisation Energy 3

Essential Pre-Uni Chemistry D3.3

Subject & topics: Chemistry | Inorganic | Periodic Table **Stage & difficulty:** A Level P2

An element has its first to fifth ionisation energies in $\mathrm{kJ\,mol^{-1}}$ listed as: $578, 1817, 2745, 11578, 14831$.

Give the group number in the periodic table that corresponds to this element.

- 15
- 17
- 14
- 16
- 13
- 18
- 1
- 2

Question deck:
STEM SMART Chemistry Week 9